

Common distributions: reference and practice

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About this sheet. This is a reference-and-practice sheet for some standard probability distributions. Each distribution is defined by its distribution function. The short exercises ask you to find the corresponding PMF or PDF and to notice a few basic properties. **This sheet will be expanded as the course progresses.**

Notation. For a discrete distribution, the probability mass function is obtained from the jumps of the DF. For a continuous distribution, the density is obtained by differentiation. The parameter choices used here will be the conventions for MAL6150 unless stated otherwise.

How to read this sheet. Each item below gives the distribution function (DF) of a named distribution. For a discrete distribution, find the probability mass function from the jumps of the DF. For a continuous distribution, find the density by differentiation. Here $\lfloor x \rfloor$ is the greatest integer not exceeding x .

- 1. Bernoulli distribution.** This models one trial with two outcomes, usually called success and failure. We write success as 1 and failure as 0. Let $0 \leq p \leq 1$, where p is the probability of success. The DF is

$$F(x) = \begin{cases} 0, & x < 0, \\ 1 - p, & 0 \leq x < 1, \\ 1, & x \geq 1. \end{cases}$$

- (a) Find $\mathbb{P}(X = 0)$ and $\mathbb{P}(X = 1)$.
- (b) What is the support of X ?
- (c) Sketch F when $p = \frac{1}{2}$.

- 2. Binomial distribution.** This models the number of successes in n independent trials, when each trial has success probability p . Let n be a positive integer and $0 \leq p \leq 1$. The DF is

$$F(x) = \begin{cases} 0, & x < 0, \\ \sum_{k=0}^{\lfloor x \rfloor} \binom{n}{k} p^k (1-p)^{n-k}, & 0 \leq x < n, \\ 1, & x \geq n. \end{cases}$$

- (a) Find the PMF of X from this DF.
- (b) Show that the case $n = 1$ gives the Bernoulli distribution.
- (c) For integers $0 \leq a \leq b \leq n$, write $\mathbb{P}(a \leq X \leq b)$ as a finite sum.

- 3. Poisson distribution.** This is often used to model the number of events in a fixed time or space interval when events occur at a constant average rate. Let $\lambda > 0$. The DF is

$$F(x) = \begin{cases} 0, & x < 0, \\ e^{-\lambda} \sum_{k=0}^{\lfloor x \rfloor} \frac{\lambda^k}{k!}, & x \geq 0. \end{cases}$$

- (a) Find the PMF of X .
- (b) Use the power series for e^λ to verify that the total probability is 1.
- (c) If $p_k = \mathbb{P}(X = k)$, show that $p_{k+1}/p_k = \lambda/(k+1)$.

- 4. Geometric distribution.** This models the number of failures before the first success in repeated independent trials, each with success probability p . Let $0 < p < 1$. The DF is

$$F(x) = \begin{cases} 0, & x < 0, \\ 1 - (1 - p)^{\lfloor x \rfloor + 1}, & x \geq 0. \end{cases}$$

- (a) Find the PMF of X .
- (b) Show that $\mathbb{P}(X \geq n) = (1 - p)^n$ for integers $n \geq 0$.
- (c) Show that for integers $m, n \geq 0$,

$$\mathbb{P}(X \geq m + n \mid X \geq m) = \mathbb{P}(X \geq n).$$

- 5. Uniform distribution.** This models a value chosen from an interval so that intervals of equal length have equal probability. Let $a < b$. The uniform distribution on $[a, b]$ has DF

$$F(x) = \begin{cases} 0, & x < a, \\ \frac{x - a}{b - a}, & a \leq x < b, \\ 1, & x \geq b. \end{cases}$$

- (a) Find its density.
- (b) For $a \leq c < d \leq b$, find $\mathbb{P}(c < X \leq d)$.
- (c) Explain geometrically why the answer in part (b) depends only on the length $d - c$.

- 6. Exponential distribution.** This is often used to model a waiting time when events occur at a constant rate λ . Let $\lambda > 0$. The DF is

$$F(x) = \begin{cases} 0, & x < 0, \\ 1 - e^{-\lambda x}, & x \geq 0. \end{cases}$$

- (a) Find its density and $\mathbb{P}(X > s)$ for $s \geq 0$.
- (b) Show that, for $s, t \geq 0$,

$$\mathbb{P}(X > s + t \mid X > s) = \mathbb{P}(X > t).$$

This is called the *memoryless property*.

- 7. Normal distribution.** This is often used for measurements that cluster around a central value μ ; the parameter σ controls the spread. Let $\mu \in \mathbb{R}$ and $\sigma > 0$. The normal distribution with parameters μ, σ^2 has DF

$$F_{\mu, \sigma}(x) = \int_{-\infty}^x \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(t - \mu)^2}{2\sigma^2}\right) dt.$$

Write $\Phi = F_{0,1}$ for the standard normal DF.

- (a) Identify the density.
- (b) By a change of variable, show that

$$F_{\mu, \sigma}(x) = \Phi\left(\frac{x - \mu}{\sigma}\right).$$

- (c) Use symmetry of the density to show that $\Phi(-x) = 1 - \Phi(x)$.